

COVID-19

COVID-19 (Coronavirus Disease 2019) is a contagious respiratory illness caused by the SARS-CoV-2 virus. First identified in late 2019, the disease spread rapidly across the world and was declared a global pandemic by the World Health Organization. Although COVID-19 primarily affects the respiratory system, it can also damage the cardiovascular, nervous, digestive, and immune systems. The severity of the disease varies significantly among individuals. Some infected people remain asymptomatic, while others experience mild flu-like symptoms, and some develop severe pneumonia, respiratory failure, or multi-organ dysfunction requiring intensive medical care.

The virus spreads mainly through respiratory droplets and fine aerosols released when an infected individual coughs, sneezes, talks, or breathes. Infection can also occur through close contact with contaminated surfaces, although this route is less common. Individuals with weakened immune systems, older adults, and patients with chronic diseases such as diabetes, hypertension, heart disease, obesity, and chronic lung diseases have a greater risk of developing severe complications. Patients infected with COVID-19 commonly experience fever, persistent dry cough, sore throat, fatigue, headache, muscle pain, and loss of taste or smell. As the disease progresses, some individuals develop chest tightness, shortness of breath, and decreased oxygen saturation due to inflammation of the lungs. In severe cases, the infection may progress to acute respiratory distress syndrome (ARDS), requiring oxygen therapy or mechanical ventilation.

Chest X-ray imaging plays an important role in evaluating moderate to severe COVID-19 infection. Although mild infections may appear normal on X-ray, advanced disease typically demonstrates bilateral patchy opacities, peripheral lung infiltrates, and diffuse areas of increased opacity involving both lungs. Computed Tomography (CT) often provides greater sensitivity and commonly reveals bilateral ground-glass opacities and consolidation.

Diagnosis is established through laboratory investigations such as Reverse Transcriptase Polymerase Chain Reaction (RT-PCR), Rapid Antigen Testing, blood investigations, pulse oximetry, and imaging studies including chest X-rays or CT scans. Treatment depends upon the severity of illness. Mild infections are generally managed with adequate hydration, rest, fever-reducing medications, and home isolation. Moderate to severe disease may require hospitalization, oxygen supplementation, antiviral therapy, corticosteroids, anticoagulants, and continuous monitoring under physician supervision.

Preventive measures include vaccination, frequent hand hygiene, maintaining adequate ventilation, wearing face masks in crowded places, avoiding close contact with infected individuals, and following public health guidelines. Medical attention should be sought immediately if the patient develops persistent breathing difficulty, oxygen saturation below 94%, chest pain, confusion, bluish lips, or severe weakness.

Pneumonia

Pneumonia is an inflammatory condition of the lungs in which the alveoli, the tiny air sacs responsible for oxygen exchange, become infected and fill with fluid or pus. The disease may be caused by bacteria, viruses, fungi, or aspiration of food and liquids into the lungs. Pneumonia affects individuals of all ages but poses a greater threat to elderly patients, young children, smokers, and people with weakened immune systems or chronic illnesses.

The infection causes inflammation that interferes with normal gas exchange, leading to reduced oxygen delivery throughout the body. The severity of pneumonia ranges from mild illness that can be treated at home to life-threatening infections requiring intensive hospital care. Bacterial pneumonia is often characterized by rapid onset and high fever, whereas viral pneumonia usually develops more gradually.

Patients commonly present with fever, productive cough, chills, chest pain that worsens during breathing, fatigue, rapid breathing, and shortness of breath. Severe pneumonia may also result in low blood oxygen levels, confusion, cyanosis, and respiratory failure if left untreated.

Chest X-ray examination remains one of the most important diagnostic tools for pneumonia. Typical findings include focal or diffuse lung consolidation, white infiltrative patches, air bronchograms, and occasionally pleural effusion. The location and extent of consolidation often help physicians determine the severity and possible cause of infection.

Diagnosis is based on clinical examination, chest X-ray findings, blood investigations, sputum culture, and pulse oximetry. Treatment depends upon the underlying organism. Bacterial pneumonia is managed with appropriate antibiotics, whereas viral pneumonia usually requires supportive care along with antiviral medications when indicated. Patients are advised to maintain adequate hydration, sufficient rest, nutritional support, and fever control. Severe cases may require hospitalization, oxygen therapy, or mechanical ventilation.

Vaccination against pneumococcal disease and influenza, maintaining good hygiene, avoiding smoking, and seeking early treatment for respiratory infections significantly reduce the risk of developing pneumonia. Immediate medical attention should be obtained if the patient experiences

severe breathing difficulty, persistent high fever, chest pain, confusion, or rapidly worsening respiratory symptoms.

Tuberculosis (TB)

Tuberculosis (TB) is a chronic infectious disease caused by the bacterium *Mycobacterium tuberculosis*. The disease most commonly affects the lungs, a condition known as pulmonary tuberculosis, although it can also involve the lymph nodes, bones, kidneys, brain, and other organs. TB remains one of the leading infectious causes of death worldwide despite the availability of effective treatment.

The disease spreads through airborne droplets released when an infected person coughs, sneezes, speaks, or sings. Prolonged exposure to an infected individual significantly increases the risk of transmission. People living with HIV/AIDS, diabetes mellitus, malnutrition, smoking habits, alcohol dependence, or weakened immune systems are at increased risk of developing active tuberculosis.

Pulmonary tuberculosis usually develops gradually over several weeks or months. The most common symptom is a persistent cough lasting longer than two weeks. Patients may also experience coughing up blood, chest pain, prolonged fever, night sweats, fatigue, reduced appetite, and progressive weight loss. Because the disease develops slowly, many patients ignore symptoms until significant lung damage has occurred.

Chest X-ray findings vary according to disease stage. Common radiological features include upper lobe infiltrates, cavitary lesions, pulmonary fibrosis, calcified granulomas, nodules, and enlarged hilar lymph nodes. Advanced disease often demonstrates extensive destruction of lung tissue.

Confirmation of tuberculosis requires laboratory testing in addition to imaging. Diagnostic investigations include sputum smear microscopy, GeneXpert molecular testing, mycobacterial culture, tuberculin skin testing, and chest radiography. Early diagnosis is essential to prevent disease progression and reduce transmission within the community.

Treatment consists of a carefully supervised combination of anti-tuberculosis medications including Isoniazid, Rifampicin, Pyrazinamide, and Ethambutol for a minimum duration of six months. Completing the full course of medication is critical because incomplete treatment may lead to multidrug-resistant tuberculosis, which is considerably more difficult to cure.

Preventive measures include BCG vaccination in childhood, early detection of infected individuals, adequate ventilation, respiratory hygiene, and completion of prescribed treatment. Medical evaluation should be sought whenever a person develops a persistent cough lasting more than two weeks, unexplained weight loss, prolonged fever, night sweats, or coughing of blood.

Normal Chest X-ray

A normal chest X-ray represents the healthy appearance of the thoracic cavity without evidence of significant disease or structural abnormality. It serves as an important baseline for comparison during the evaluation of respiratory and cardiovascular disorders. A properly exposed normal chest X-ray demonstrates clear lung fields without infiltrates or consolidation, normal heart size, intact diaphragms, well-defined costophrenic angles, normal mediastinal contours, and no evidence of pleural effusion, pneumothorax, masses, or fractures.

The lungs should appear well expanded with symmetrical vascular markings and no abnormal opacities. The heart should occupy less than half the width of the thoracic cavity on a standard posteroanterior chest radiograph. The diaphragm should have smooth contours, and both costophrenic angles should remain sharp and free of fluid accumulation. Although a normal chest X-ray is reassuring, it does not completely exclude disease because very early infections, small tumors, or subtle abnormalities may not yet be visible. Therefore, physicians always interpret imaging findings together with the patient's medical history, physical examination, laboratory investigations, and, when necessary, advanced imaging such as CT scans.

Diabetes Mellitus

Diabetes Mellitus is a chronic metabolic disorder characterized by persistently elevated blood glucose levels resulting from inadequate insulin production, impaired insulin action, or both. Insulin is a hormone produced by the pancreas that enables glucose to enter body cells, where it is used as the primary source of energy. When insulin is insufficient or ineffective, glucose accumulates in the bloodstream, leading to hyperglycemia and long-term damage to multiple organ systems.

There are three principal forms of diabetes. Type 1 Diabetes results from autoimmune destruction of insulin-producing pancreatic beta cells and usually develops during childhood or adolescence, requiring lifelong insulin therapy. Type 2 Diabetes, the most common form, develops when body tissues become resistant to insulin and is strongly associated with obesity, physical inactivity, unhealthy dietary habits, and genetic predisposition. Gestational Diabetes occurs during

pregnancy and usually resolves after delivery, although affected women remain at increased risk of developing Type 2 Diabetes later in life.

Patients with diabetes commonly experience excessive thirst, frequent urination, increased hunger, unexplained weight loss, fatigue, blurred vision, slow wound healing, recurrent skin or urinary infections, and numbness or tingling in the hands and feet. Poorly controlled diabetes can lead to serious complications including heart disease, stroke, kidney failure, diabetic retinopathy causing blindness, peripheral neuropathy, diabetic foot ulcers, and increased susceptibility to infections.

Diagnosis is established using laboratory investigations such as fasting blood glucose, random blood glucose, oral glucose tolerance testing, and glycated hemoglobin (HbA1c), which reflects average blood sugar levels over approximately three months. Effective management involves lifestyle modification, a balanced diet, regular physical activity, weight reduction when appropriate, routine blood glucose monitoring, and the use of oral antidiabetic medications or insulin therapy as prescribed by a physician.

Individuals should seek immediate medical attention if they experience extremely high blood glucose levels, persistent vomiting, severe dehydration, confusion, rapid breathing, chest pain, or symptoms suggestive of diabetic ketoacidosis or hyperosmolar hyperglycemic state, both of which are life-threatening emergencies.